

Welcome to Code Crunch!

C12 | Crunch 3D: CAD & Modeling

[Tues 4-5 PM] Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



CODECRUNCH.CS.FIU



Unit #1

Topic: Introduction & Sketching
C12 | Crunch 3D: CAD & Modeling



What Code Crunch Offers

- **Hands-On Workshops:** Topics include Python, JavaScript, React, and more.
- **LeetCode Prep:** Weekly sessions to master technical interviews.
- **Hackathons:** Collaborate on innovative projects in a competitive setting.
- **Panel Discussions:** Industry leaders from tech giants and startups share insights.
- **Global Access:** Code Crunch Global launches in October for worldwide participation!

305 Hackathon : Edition Spring 2025





OPEN TO ALL STUDENTS, REGARDLESS OF MAJOR
—NO EXPERIENCE NEEDED! FRESHMEN,
GRADUATES, AND ALUMNI FROM ANY UNIVERSITY
ARE WELCOME. LET'S INNOVATE TOGETHER!

Funded by SGA
| Open to All University Students

305



SPRING 2025

HACKATHON

HYBRID

MAR 24 - 27 & MAR 29 - 31, 2025
WORKSHOPS | COMPETE | IN-PERSON CAREER FAIR

FIU
Honors College

epichire

SIGN UP HERE: [CODECRUNCH.CS.FIU.EDU](https://codecrunch.cs.fiu.edu)

FIU ORGS COLLABORATORS





#CODECRUNCH-HACKS

Spring 2025 Weekly Schedule





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CODE CRUNCH WEEKLY SCHEDULE - SPRING 25

Your CS Club for Collaboration, Networking, and Hands-On Projects!
Open to all students, regardless of major—no experience needed! **Freshmen, graduates, and alumni from any university are welcome.** Let's innovate together!

C5 Crunch AI Data Science Python MON 1 - 2 PM WED 12:30 - 1:30 PM	C3 CodeCrunch LABS Portfolio Java & React MON 2 - 3 PM	C7 Wire Crunch: Embedded Systems C++ TUE 1 - 3 PM	C11 Crunch Arcade Unity TUE 3 - 4 PM FRI 4 - 5 PM	C12 Crunch 3D: CAD & Modeling OnShape TUE 4 - 5 PM	C9 Swe Convos Networking WED 1:30 - 2:30 PM FRI 2 - 3PM ***
C1 Crunch Convos Python WED 2:30 - 3:30 PM	C2 Crunch The Code Python WED 3:30 - 4:30 PM	C6 Cybersecurity Crunch VM/LINUX THU 1 - 2 PM	C4 CodeCrunch LABS Hackathon Java & React THU 2 - 3 PM	C8 CodeCrunch LABS: Web Dev Javascript THU 3 - 4 PM	C10 Crunching Bioinformatics Python FRI 3 - 4 PM

Attend 7 of 10 Weekly Sessions in the Same Workshop to Earn a Certificate of Completion!

RSVP on
LU.MA/CODECRUNCH

In person or
Stream on
Zoom



 CODECRUNCH.CS.FIU.EDU

 **January 6, 2025 - April 4, 2025**
Cost: No-cost

Become a member



What we offer at C12

Crunch 3D: CAD & Modeling



- This **6-week** course introduces the basics of **On-Shape**, a **3D CAD** software used in engineering, product design, and manufacturing.
- We will **guide** you through the **essential skills** needed to bring your ideas to life, including **sketching**, **3D modeling**, and **assemblies**.
- By the **end** of this course, you will not only have a **finished project** to showcase but also a **solid understanding** of On-Shape.
- Whether you dream of **creating** your own 3D models, **breaking** into the **3D design industry**, or **simply exploring** your creativity, this course will give you the foundation you need to get started!

Key Dates and Structure

C12 - Crunch 3D: CAD & Modeling



★ **Week 1: Introduction to OnShape, Sketching Fundamentals, & 2D Constraints**

Learning what On-Shape is and how to properly sketch on OnShape.

★ **Week 2: 3D Part Modeling**

Learning how to use extrude, revolve, fillets, chamfers, and functions for 3D models.

★ **Week 3: 3D Part Modeling Continued**

Learning how to use sweep, loft, shell, rib, and mirror functions for 3D models.

★ **Week 4: Assemblies & Mating Constraints**

Learning how to create assemblies through mates and constraints.

★ **Week 5: Final Project - Real-World Design Challenge**

Applying all concepts to design a functional product.

★ **Week 6: Final Project Showcase**

Presenting your final project, sharing your experience, and receiving feedback.



Starting off...

Let's Get Started!



onshape®



OnShape Overview

- OnShape is a 3D CAD modeling software that is **free!**
- OnShape is accessible through a **web browser** as well as a **mobile app**.
- With OnShape you can **share and collaborate** on 3D models live (you can edit the same file at the same time!).
- OnShape allows for **3D modeling, assemblies, simulations, and engineering drawings**.



Creating an Account

1. Search “Sign up for OnShape” or click <https://www.onshape.com/en/sign-up>
2. Fill out the information (any name, any last name, any email!)
3. For use, select either Education or Fun
4. Continue with the steps to fill out your account information
5. You now have access to OnShape!



What is sketching?

- **Sketching** is a 2D drawing feature which allows you make the base for your 3d design.
- **Sketch entities** are the lines or curves drawn in a sketch.
- **Dimensions** are the lengths you give to sketch entities created in the sketch.
- **Constraints** are limitations created by the dimension given to different sketch entities.
- **Relations** are relationships between sketch entities (tangent, parallel, etc.).



Time for...

Sketching!



Q&A Session



Attendance



Join us for the upcoming weekly sessions this Spring 2025!
(Jan 6 - Apr 4, 2025)



Monday: C5 1PM
Tuesdays: C7 1PM / C11 3PM
Wednesdays: C5 12:30 PM / C9 1:30 PM (Varies)
C1 2:30PM / C2 3:30PM / C13 5:30PM
Thursdays: C6 1PM / C8 3PM
Friday: C9 2PM (Varies) / C10 3PM / C11 4PM



RSVP [LU.MA/CODECRUNCH](https://lu.ma/codecrunch)



Your feedback helps us improve. Share your thoughts!
Go to our website: [CODECRUNCH.CS.FIU](https://codecrunch.cs.fiu)